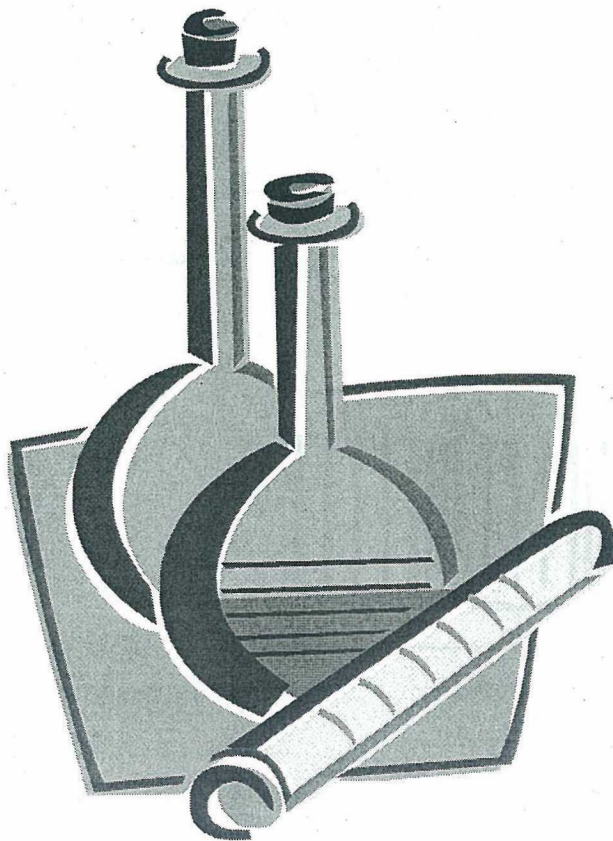


YR. 10

SOUND
LIGHT
GENETICS

SCIENCE

**EXAM REVISION
BOOKLET**



Chapter 2

Use with Section 1

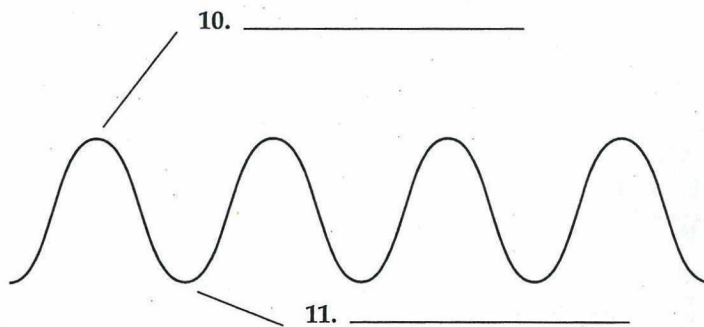
REINFORCEMENT**• What are waves?**

Write the kind or kinds of waves—mechanical, electromagnetic, transverse, or compressional—that apply to each term or description.

1. X rays _____
2. water waves _____
3. can travel only through a medium _____
4. transfers energy but not matter _____
5. has peaks and valleys _____
6. matter moves in same direction as wave travels _____
7. can travel through a vacuum _____
8. has areas where particles are spread out _____
9. matter moves at right angles to direction wave travels _____

Label each diagram and give an example of each kind of wave.

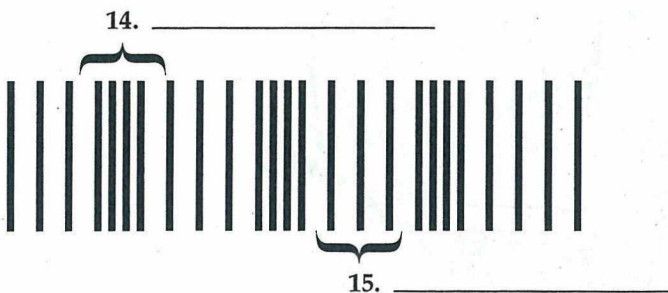
A.



12. _____ wave

13. Example: _____

B.



16. _____ wave

17. Example: _____

Chapter 2

Use with Section 2

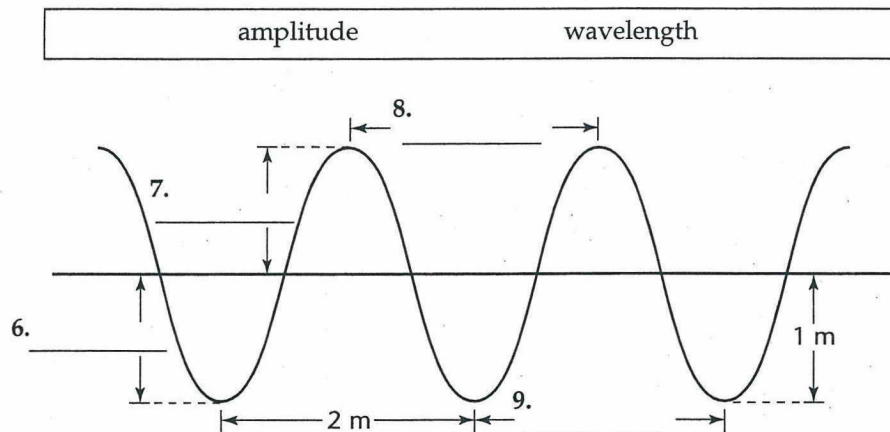
REINFORCEMENT

● Wave Properties

Complete each sentence by circling the correct answers and crossing out the incorrect answer.

1. The amplitude of a wave can be measured from the (medium, crest) or the (trough, wavelength) to the rest position of the wave's medium.
2. Waves with greater amplitudes carry (more, less) energy than waves with smaller amplitudes.
3. The wavelength of a transverse wave is often measured from (crest to crest, crest to trough).
4. The number of waves that pass a point in one (second, minute) is the wave's (amplitude, frequency).
5. Waves with longer wavelengths have a (lower, higher) frequency and waves with shorter wavelengths have a (lower, higher) frequency.

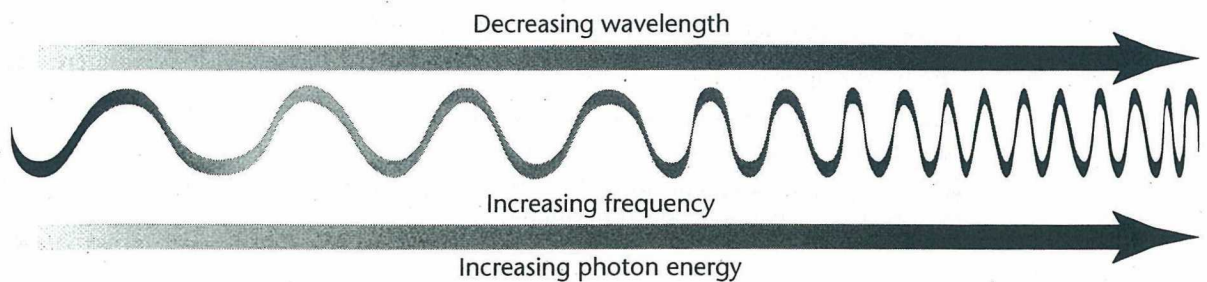
Use the words in the box to label the diagram. You will use each term more than once. Then answer the questions.



10. What is the wavelength of the wave shown in the diagram? _____
11. What is the amplitude of the wave shown in the diagram? _____

Chapter 3

Use with Section 1

ENRICHMENT● **Properties of Light****Creatures of the Spectrum**

Gamma rays	X rays	Ultraviolet	Visible	Infrared	Microwave	Radio
used to kill cancer cells	used to examine bones and teeth; also used to "see" inside luggage for airport security	used to sterilize food and medical equipment	light we can see	used to cook food and for special "night vision" cameras and binoculars, which can "see" in the dark	used for cooking food and for radar	used for communication

Visible light is only a small part of the electromagnetic spectrum. It is important to humans because human eyes can see only that part of the spectrum. Other animals, however, see other sections of the spectrum. To humans, for instance, a daisy has a gold center surrounded by rays of white petals. Honeybees, however, see ultraviolet light, and when they look at a daisy, they see ultraviolet patterns that circle the flower's center. To the honeybee, the simple daisy looks like a bull's-eye.

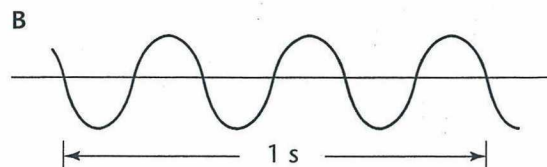
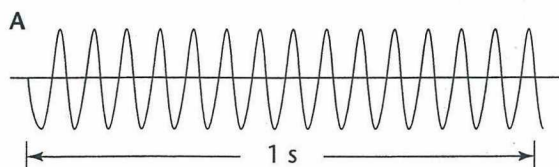
If creatures live elsewhere in the universe, they may be able to "see" different parts of the electromagnetic spectrum than earthlings. Imagine that you have discovered a new life-form on another planet and have discovered that it "sees" a different band of the spectrum than humans. Using the spectrum information, describe your otherworldly creatures in the space below.

Chapter 3

Use with Section 1

REINFORCEMENT**● Properties of Light**

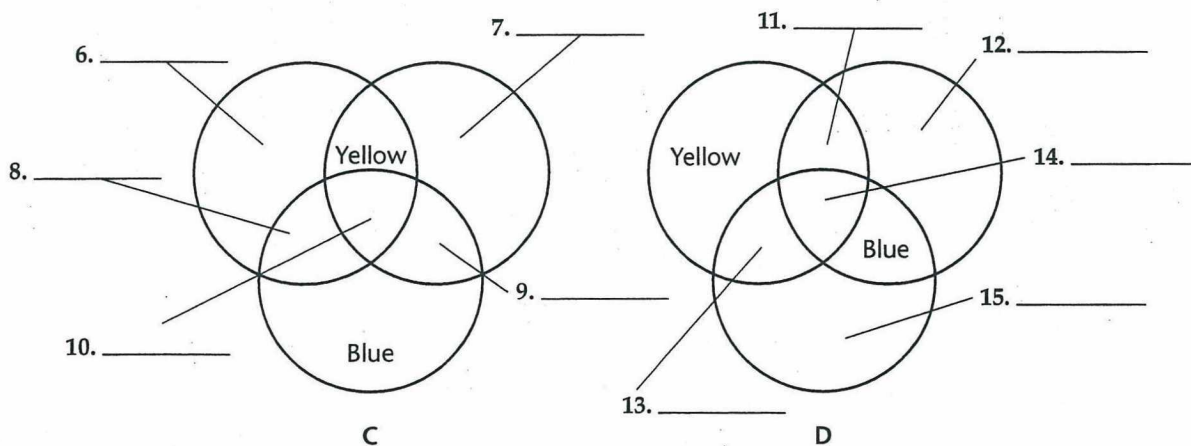
Use the light waves below and a metric ruler to answer the questions that follow.



1. What is the wavelength of wave B? _____
2. What is the frequency of wave A? _____
3. What is the frequency of wave B? _____
4. Which wave carries more energy? Why? _____

5. Which wave could represent a radio wave? Why? _____

Complete the charts by filling in the missing colors and answer the questions.



16. Which chart shows what happens when pigments are mixed? _____
17. What colors of light are reflected by an object that appears white? _____

Answer the following questions.

18. Why can't you see a book in a dark room? _____
19. How is an opaque object different from a translucent object? _____

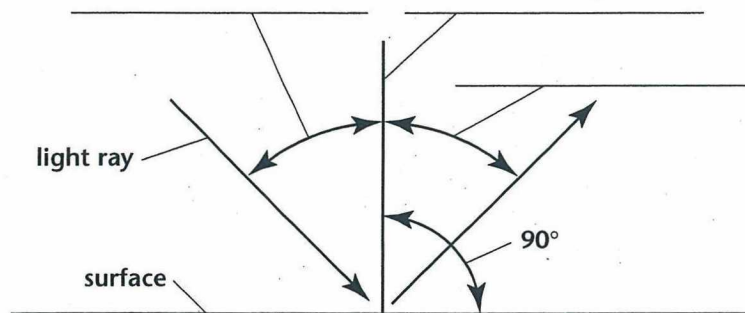
Chapter 3

Use with Section 2

REINFORCEMENT**● Reflection and Mirrors**

Use items from the box to complete the diagram. Then complete the sentence that describes the diagram.

angle of incidence
angle of reflection
the normal



The *law of reflection* states that the _____ of _____ is equal to the _____ of _____.

Complete each item.

1. Light rays reflect off a rough surface.
 - a. Do the rays reflect in many directions or few? _____
 - b. What type of reflection is this? _____
2. Light rays reflect off a smooth surface.
 - a. Do the rays reflect in many directions or few? _____
 - b. What type of reflection is this? _____
3. A mirror has a flat reflecting surface. What kind of mirror is this? _____
4. A mirror has a reflecting surface that is curved inward. What kind of mirror is this?

5. A mirror has a reflecting surface that is curved outward. What kind of mirror is this?

Label each diagram as a plane mirror, a concave mirror, or a convex mirror.



Chapter 3

Use with Section 3

REINFORCEMENT**● Refraction and Lenses**

1.–5. Read each pair of statements. One or both of them are correct. Circle the ones that are correct. Cross out any incorrect ones.

Light travels at the same speed in all mediums.

Light travels at different speeds in different mediums.

Refraction is the change of speed of a light wave as it moves from one medium to another.

Refraction is the bending of a light wave as it moves from one medium to another.

The larger the change in the speed of a light wave, the greater its refraction.

The larger the change in the speed of a light wave, the lesser its refraction.

A lens is a transparent object with at least one curved side that causes light to refract.

A lens is a transparent object with two curved sides that cause light to refract.

A convex lens is also called a converging lens.

A concave lens is also called a diverging lens.

Identify each statement as describing a convex lens, a concave lens, or both.

- | | |
|-------|---|
| _____ | 6. A lens that is thicker in the center than at the edges |
| _____ | 7. A lens that is thicker at the edges than in the center |
| _____ | 8. Causes light rays passing through it to meet at a focal point |
| _____ | 9. Has an optical axis |
| _____ | 10. An object more than two focal lengths from the lens will have an inverted image |
| _____ | 11. Causes light rays passing through it to diverge |
| _____ | 12. Causes light rays passing through it to refract |
| _____ | 13. Used to correct nearsightedness |
| _____ | 14. Used to correct farsightedness |
| _____ | 15. Creates a focal point |

Chapter 6

Use with Section 3

ENRICHMENT**• Electric Circuits****To Resist or Not to Resist**

Georg Ohm was a German physicist born in 1787. He is most famous for his work studying the resistance of materials to electric current. The unit of measure for electrical resistance is named for him, the ohm. Resistance is defined as the ratio of the potential difference, V , to the current, I .

$$R = V/I$$

One ohm is equal to the resistance that will allow 1 ampere (A) (the electrical current) to flow through 1 volt (V) (the potential difference).

Ohm's law states that most conductors' resistance is not determined by the size or direction of the potential difference across it. Any device that maintains a constant resistance, as long as the temperature does not change, obeys Ohm's law. Fortunately, not all conductors obey Ohm's law. Transistors provide one example of a device that does not. Regardless of whether a device follows Ohm's law, the equation $R = V/I$ can still be used.

Apply what you've learned.

1. What is the current through a 40-ohm resistance that has a potential difference of 160 volts?

2. If the headlight on your mom's car has a resistance of 32 ohms and the car battery put out 12 V, what is the current through the headlight circuit?

3. The motor for your electric car is attached to a generator. The motor has a resistance of 28 ohms and the current is 3.8 amperes (A). What is the generator's voltage?

Challenge question.

If a clock radio uses 2×10^{-4} A of current when you run it on a 3-V battery, what resistance is its circuit providing?

Chapter 6

Use with Section 3

REINFORCEMENT**● Electric Circuits***Use the terms and statements from this box to complete the table.*

amount of electric energy used by a device	a circuit that has more than one path for the electric current to follow
a circuit that has only one path for the electric current to follow	George Simon Ohm
kilowatt	kW
Ohm's law	$P = I \times V$
parallel	parallel circuit
power = current $\times$ voltage	series
series circuit	$V = I \times R$
voltage = current $\times$ resistance	W
watt	

IMPORTANT FACTS ABOUT ELECTRIC CIRCUITS**1. There is a relationship among voltage, current, and resistance in an electric circuit.**

Name of law:	
Discovered by:	
Expression of law:	
Equation:	

2. There are two types of electric circuits.

Two types of circuits:	1. 2.
Definitions of these circuits	1. 2.

3. The electrical power of a circuit can be measured.

Definition of electrical power:	
Unit of electrical power:	1. Name: 2. Abbreviation: 3. Term for 1000 units: 4. Abbreviation for 1000 units:
Determining the electrical power of a circuit:	1. Expression: 2. Formula:

Chapter 6

Use with Section 2

REINFORCEMENT**• Electric Current**

Complete the paragraphs using words from the box.

chemical reactions

dry-cell

electric potential energy

ohms

resistance

volts

circuit

electric current

negative

positive

V

wet-cell

Life as we know it would be impossible without electricity. Think of the number of electrical devices we rely on every day: lights, refrigerators, computers, televisions, flashlights, car headlights, watches—the list is endless. All of these devices, and countless others, need a constant, steady source of electrical energy. This steady source of electrical energy comes from a(n) _____, which is the steady flow of electrons through a conductor.

This steady flow of electricity requires a closed path, or _____, through which to flow. Its basic elements are a conductor, such as wire, through which electrons flow and a source of electrons, such as a battery.

An electric current carries energy that comes from separating positive and negative charges. Negatively charged electrons “seek out” positively charged electrons to recombine. This can only happen if they travel through the circuit. In a circuit, the electrons flow from the _____ end to the _____ end.

A familiar source of electrons in electric circuits is a battery. The total stored electrical energy in a battery—the energy available to do work—is called _____. This energy is measured in units called _____, which is abbreviated _____. Two common types of batteries are _____ and _____ batteries. Both rely on _____ to separate positive and negative electrical charges. When the negative and positive ends of the charges are connected by a conductor, a circuit forms and the electrical energy is available to do work.

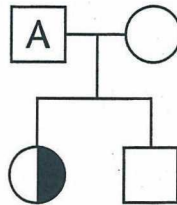
However, the electrons don’t flow completely freely through the circuit. Depending on the material used for the conductor, the electrons have more or less difficulty flowing. The measure of how difficult it is for electrons to flow through a circuit is called _____. This is measured in units called _____.

Chapter 17

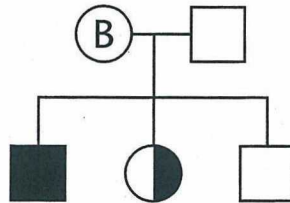
Use with Section 3

REINFORCEMENT**• Human Genetics****Part A**

Each of the pedigrees below shows the inheritance pattern of a sex-linked disorder, such as color blindness. Answer the question below each pedigree.



1. Is the father, A, affected or not affected by the disorder? Explain your answer. _____



2. Is the mother, B, affected by the disorder, not affected, or a carrier? Explain your answer.

Part B

1. Name two genetic disorders that are caused by a recessive allele on the X chromosome.

2. Name two genetic disorders that are homozygous recessive. _____

3. How are scientists in the field of genetic engineering trying to help people with genetic disorders? _____

Chapter 17

Use with Section 3

ENRICHMENT

● Human Genetics

Pedigrees

Consider a sex-linked trait such as color blindness, inherited on the X chromosome. Let C stand for the dominant or normal allele and c for the recessive allele responsible for color blindness. Only a female has two alleles for the trait. A male inherits either a C or c allele along with the X chromosome from the mother. A male also inherits a Y chromosome from the father, which has no allele for color blindness. Consequently, a female's genotype can be CC, cc, or Cc, and a male's genotype can be YC or Yc. Remember that a male who has the recessive allele is affected by color blindness, while a female must have two recessive alleles to be affected. A heterozygous female is a carrier of the trait, but is not affected by it.

Make a Punnett square to show the possible offspring of a carrier female and a normal male.

	C	c
Y		
C		

Draw a pedigree for this family. Assume that there are four children, and that they inherit the trait in the ratio predicted by the Punnett square.

Assume that the homozygous daughter marries a color-blind male and has a daughter. Add the husband and daughter to the pedigree. Be sure to indicate whether the daughter is affected, not affected, or a carrier.

Oompah Loompa Genetics

Mendelian Genetics and Codominance

SHOW ALL WORK and PUNNET SQUARES

1. Oompahs generally have blue faces which is caused by a dominant gene. The recessive condition results in an orange face. Develop a "key" to show the genotypes and phenotypes possible for Oompa Loompas.
2. Two heterozygous OOmpahs are crossed. What proportion of the offspring will have orange faces.
3. A blue faced Oompah (homozygous) is married to an orange faced Oompah. They have 8 children. How many children will have blue faces?
4. Otis Oompah has an orange face and is married to Ona Oompah who has a blue face. They have 60 children, 31 of them have orange faces. What are the genotypes of the parents.
5. Odie Oompah has a blue face. In fact, everyone in Odie's family has a blue face, and the family boasts that it is a "pure" line. Much to his family's horror, he married Ondi Oompah who "gasp" has an orange face. What are the genotypes of their children. Is Odie's line still "pure"?
6. Ona Oompah (from #4) divorces Otis and marries Otto. Otto has an orange face. What is the probability that Ona and Otto's children will have orange faces?
7. Oompahs can have red, blue or purple hair. Purple hair results from the heterozygous condition. Make a "key" showing the genotypes and phenotypes for hair color. Is this an example of codominance or incomplete dominance?
8. Orville Oompah has purple hair and is married to Opal Oompah who brags that she has the bluest hair in the valley. How many of Opal's children will be able to brag about their blue hair also?
9. One of Opal's children is born with shocking red hair. Is Orville the father of this child (show the square to prove

your answer)? But wait, Opal swears she has been faithful and claims that the hospital goofed and got her baby mixed with another. Is this a plausible explanation. Show the square to prove your answer.

10. Olga Oompah has red hair and marries Oliver Oompah who has blue hair. They have 32 children. What color is their children's hair?

11. Olivia Oompah is married to Odo Oompah and they both have purple hair. What color hair and in what proportion would you expect their children to have?

12. In the land of Oompah, blue hair is highly valued, blue haired Oompahs even get special benefits. Oscar Oompah has purple hair but he wants to find a wife that will give him blue haired children. What color hair should his wife have? What would be his second choice?

BIOLOGICAL EVIDENCE – COMPARATIVE EMBRYOLOGY

- ◆ Comparison of embryos of terrestrial species (e.g. humans, chickens, pigs) show pharyngeal _____ that resemble the gill slits of fish. Also these embryos show _____ in the early gestation period, slowly degenerating before the end of the gestation period.

BIOLOGICAL EVIDENCE – VESTIGIAL STRUCTURES

- ◆ Some species have structures that do not _____ at present, but may have functioned in an ancestor (e.g. appendix and _____ teeth in humans).

BIOLOGICAL EVIDENCE – HOMOLOGOUS STRUCTURES

- ◆ **Analogous structures** are anatomical structures that are found in many unrelated organisms. For example, wings of birds and _____ serve the same function of flight. These similar structures do _____ show signs of an evolutionary relationship.
- ◆ **Homologous structures** are structures that are similar (e.g. the pentadactyl or five-fingered structure of a human _____ compared with the five-fingered structure of a _____) and also show signs of an evolutionary relationship.

BIOLOGICAL EVIDENCE – PEPPERED MOTH

- ◆ This is observed example of Natural _____
- ◆ Before the Industrial Revolution in England, a forest-dwelling moth (*Biston betularia*) grew in 3 shades – light, peppered and dark. As the tree bark was lighter, the most common shade of moth was the peppered one. However, after the soot from the factories darkened the barks of trees in the moth habitat, the most common moth was the _____ one. The reason was probably because the light and peppered moths were no longer _____ and were being by predators. The dark moths then reproduced, changing the gene frequencies of the population.

BIOCHEMICAL EVIDENCE – BACTERIAL RESISTANCE TO ANTIBIOTICS

- ◆ This is an observed example of Natural Selection.
- ◆ Bacteria do _____ become resistant to antibiotics. Either they carry the genetic trait for resistance or they _____
- ◆ Before World War II, there were different kinds of bacteria – some which were resistant to _____ and some which were not. After the widespread use of antibiotics during and after World War II that killed most of the non-resistant bacteria, the resistant bacteria continued to _____ even more due to lack of competition. Their genetic resistance was passed on to subsequent generations. Many of the bacteria that we encounter today are _____ to our present antibiotics.

BIOCHEMICAL EVIDENCE – MUTATIONS

- ◆ A mutation is a change in the _____ sequence of a DNA molecule or of a chromosome.
- ◆ A mutagen is a chemical or a form of _____ that can cause a mutation.
- ◆ Since DNA controls the synthesis of _____ (e.g. skin collagen, hair keratin, muscle myosin, hormones, enzymes, haemoglobin, antibodies), the proteins formed may be affected.
- ◆ Mutations can occur naturally. However, they are _____ events (about 1 in every 100000 generations).
- ◆ However, mutations of fruit flies are believed to be _____ since mutated flies seem to be in few numbers.
- ◆ One of the bases on which the theory of evolution is built is that chance mutations formed new organisms which could no longer mate with the rest of their species and eventually formed new species. However, since mutations observed in fruit fly experiments are so rare and often cause harm to the individuals, creationists argue that evolution in this way is not plausible.

LONGMAN GCSE BIOLOGY WORKSHEETS

5: Natural selection

This worksheet uses some invented data to test your understanding of the principles of natural selection. It also deals with some aspects of ecology.

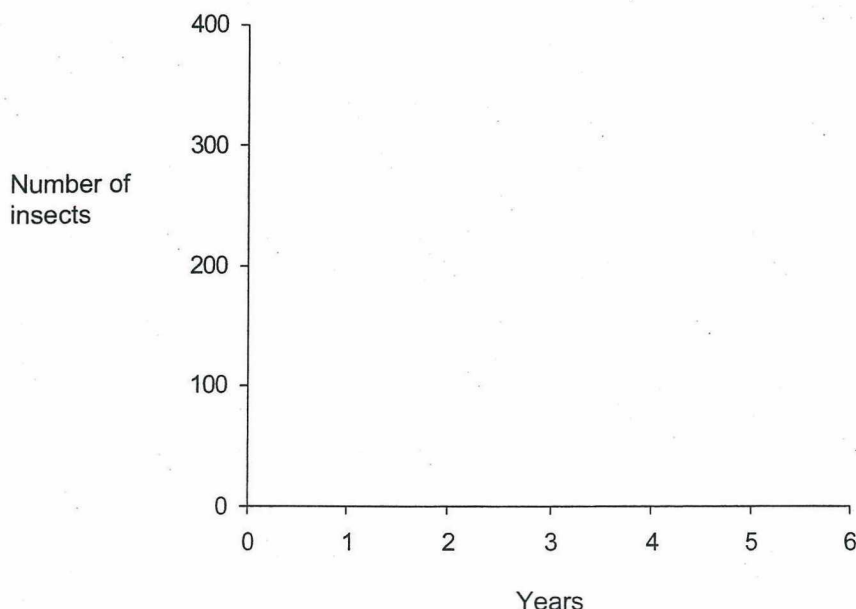
A farmer's grain crop is under attack from an insect pest. This pest, the corn weevil, bores holes in the seeds of the crop in summer and lays its eggs. The adults then die. The eggs hatch into larvae, which feed from the seeds until they mature and are ready to fly off. The adult weevils feed from a different food plant. They mate and then hibernate through the winter, starting the cycle again the next year. For every adult weevil, an average of 5 offspring are produced every year.

The farmer has tried to control the corn weevil by spraying his crops in summer with an insecticide called 'Bugloss'. When he first used this, it was very successful. Only 15% of the immature weevils survived the application of Bugloss. However, in the last year of spraying, 25% survived. The corn weevil is known to develop a mutation which makes it resistant to Bugloss, and the farmer has been advised that resistance has developed in the pest attacking his crop. This resistance is inherited.

Assuming that weevils that are resistant to Bugloss show a 25% survival rate following spraying, while non-resistant weevils show a 15% survival rate, and that both resistant and non-resistant weevils produce 5 offspring per year, answer the following questions:

Questions

- Starting with 100 insects, calculate the number of weevils you would expect to find each year for six successive years, for (a) resistant insects and (b) non-resistant insects (assuming the survival rate of the insects stays constant at 25% and 15% each year).
- a) Plot these numbers on a graph, using the same axes for both types of insect:



- Comment on the shape of the graph.
- Explain the terms **mutation** and **resistance** (to insecticide).

4. Darwin's theory says that evolution takes place as a result of the process of **natural selection**, through **survival of the fittest**. With reference to your calculations and the graph, explain how the example of the corn weevil illustrates these processes.

5. If the farmer continues to spray with bugloss, he will be wasting his money. Describe another way he could reduce the numbers of the corn weevil, without using chemical insecticides.

6. If he continues to use insecticides, there are a number of harmful side effects of these chemicals that could harm the ecosystem. Describe **three** of these.

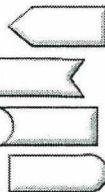
DNA - The Double Helix

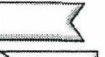
Recall that the nucleus is a small spherical, dense body in a cell. It is often called the "control center" because it controls all the activities of the cell including cell reproduction, and heredity. Chromosomes are microscopic, threadlike strands composed of the chemical DNA (short for deoxyribonucleic acid). In simple terms, DNA controls the production of proteins within the cell. These proteins in turn, form the structural units of cells and control all chemical processes within the cell. Think of proteins as the building blocks for an organism, proteins make up your skin, your hair, parts of individual cells. How you look is largely determined by the proteins that are made. The proteins that are made is determined by the sequence of DNA in the nucleus.

Chromosomes are composed of genes, which is a segment of DNA that codes for a particular protein which in turn codes for a trait. Hence you hear it commonly referred to as the gene for baldness or the gene for blue eyes. Meanwhile, DNA is the chemical that genes and chromosomes are made of. DNA is called a nucleic acid because it was first found in the nucleus. We now know that DNA is also found in organelles, the mitochondria and chloroplasts, though it is the DNA in the nucleus that actually controls the cell's workings.

In 1953, James Watson and Francis Crick established the structure of DNA. The shape of DNA is a double helix (**color the title black**), which is like a twisted ladder. The sides of the ladder are made of alternating sugar and phosphate molecules. The sugar is deoxyribose. **Color all the phosphates pink (one is labeled with a "p")**. **Color all the deoxyriboses blue (one is labeled with a "D")**.

The rungs of the ladder are pairs of 4 types of nitrogen bases. The bases are known by their coded letters A, G, T, C. These bases always bond in a certain way. Adenine will only bond to thymine. Guanine will only bond with cytosine. This is known as the "Base-Pair Rule". The bases can occur in any order along a strand of DNA. The order of these bases is the code that contains the instructions. For instance ATGCACATA would code for a different gene than AATTACGGA. A strand of DNA contains millions of bases. (For simplicity, the image only contains a few.)

Color the thymines orange. 

Color the adenines green. 

Color the guanines purple. 

Color the cytosines yellow. 

Note that the bases attach to the sides of the ladder at the sugars and not the phosphate.

The DNA helix is actually made of repeating units called nucleotides. Each nucleotide consists of three molecules: a sugar (deoxyribose), a phosphate which links the sugars together, and then one of the four bases. Two of the bases are purines - adenine and guanine. The pyrimidines are thymine and cytosine. Note that the pyrimidines are single ringed and the purines are double ringed. Color the nucleotides using the same colors as you colored them in the double helix.

The two sides of the DNA ladder are held together loosely by hydrogen bonds. The DNA can actually "unzip" when it needs to replicate - or make a copy of itself. DNA needs to copy itself when a cell divides, so that the new cells each contain a copy of the DNA. Without these instructions, the new cells wouldn't have the correct information. The hydrogen bonds are represented by small circles. **Color the hydrogen bonds grey.**

Messenger RNA

So, now, we know the nucleus controls the cell's activities through the chemical DNA, but how? It is the sequence of bases that determine which protein is to be made. The sequence is like a code that we can now interpret. The sequence determines which proteins are made and the proteins determine which activities will be performed. And that is how the nucleus is the control center of the cell. The only problem is that the DNA is too big to go through the nuclear pores. So a chemical is used to read the DNA in the nucleus. That chemical is messenger RNA. The messenger RNA (mRNA) is small enough to go through the nuclear pores. It takes the "message" of the DNA to the ribosomes and "tells them" what proteins are to be made. Recall that proteins are the body's building blocks. Imagine that the code taken to the ribosomes is telling the ribosome what is needed - like a recipe.

Messenger RNA is similar to DNA, except that it is a single strand, and it has no thymine. Instead of thymine, mRNA contains the base Uracil. In addition to that difference, mRNA has the sugar ribose instead of deoxyribose. RNA stands for **Ribonucleic Acid**. Color the mRNA as you did the DNA, except:

Color the ribose a DARKER BLUE, and the uracil brown. 

The Blueprint of Life

Every cell in your body has the same "blueprint" or the same DNA. Like the blueprints of a house tell the builders how to construct a house, the DNA "blueprint" tells the cell how to build the organism. Yet, how can a heart be so different from a brain if all the cells contain the same instructions? Although much work remains in genetics, it has become apparent that a cell has the ability to turn off most genes and only work with the genes necessary to do a job. We also know that a lot of DNA apparently is nonsense and codes for nothing. These regions of DNA that do not code for proteins are called "introns", or sometimes "junk DNA". The sections of DNA that do actually code from proteins are called "exons".

Name _____

1. Write out the full name for DNA. _____
2. What is a gene? _____
3. Where in the cell are chromosomes located?

4. DNA can be found in what two organelles?

5. What two scientists established the structure of DNA?

6. What is the shape of DNA? _____
7. What are the sides of the DNA ladder made of? _____
8. What are the "rungs" of the DNA ladder made of?

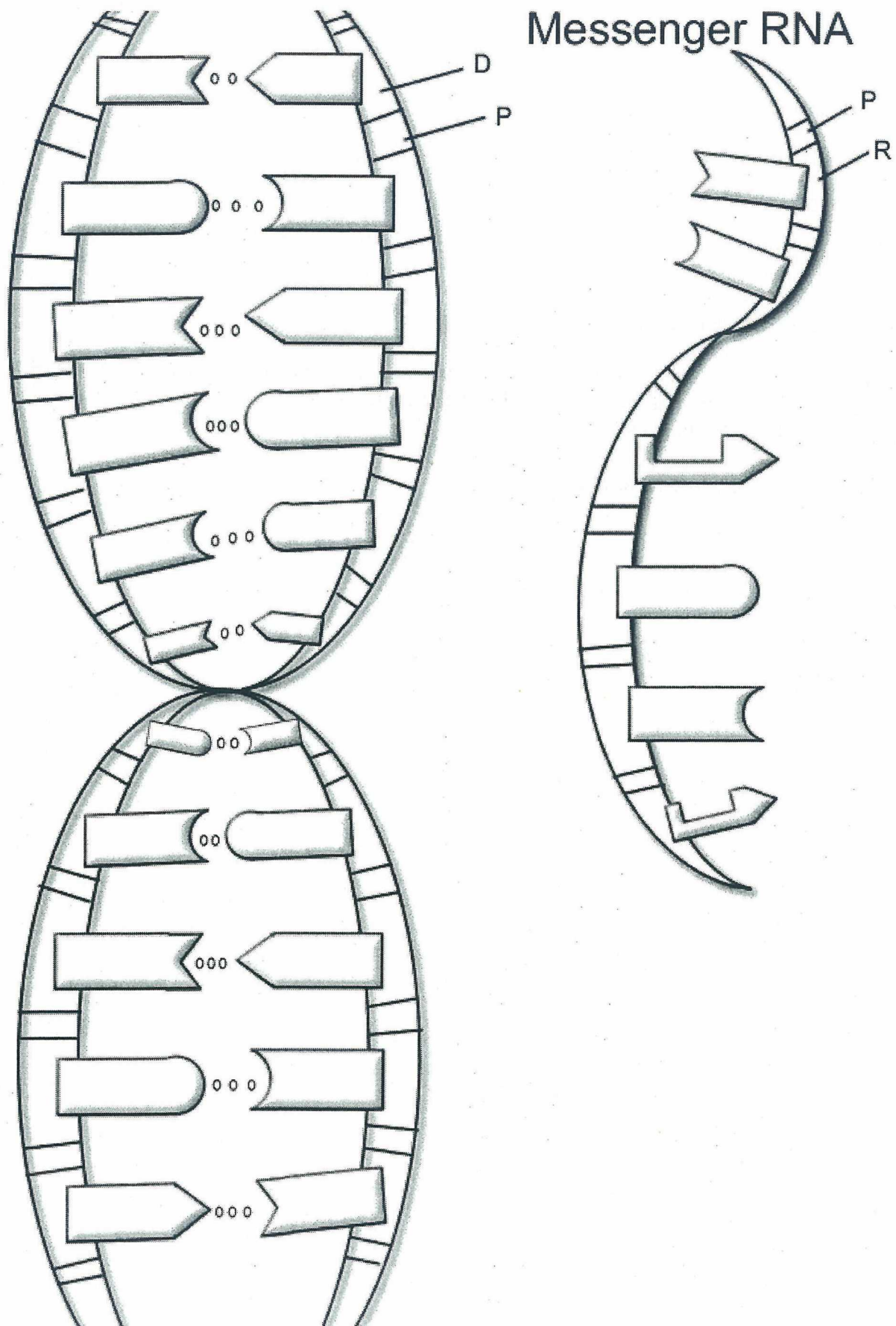
9. What sugar is found in DNA? _____ In RNA? _____

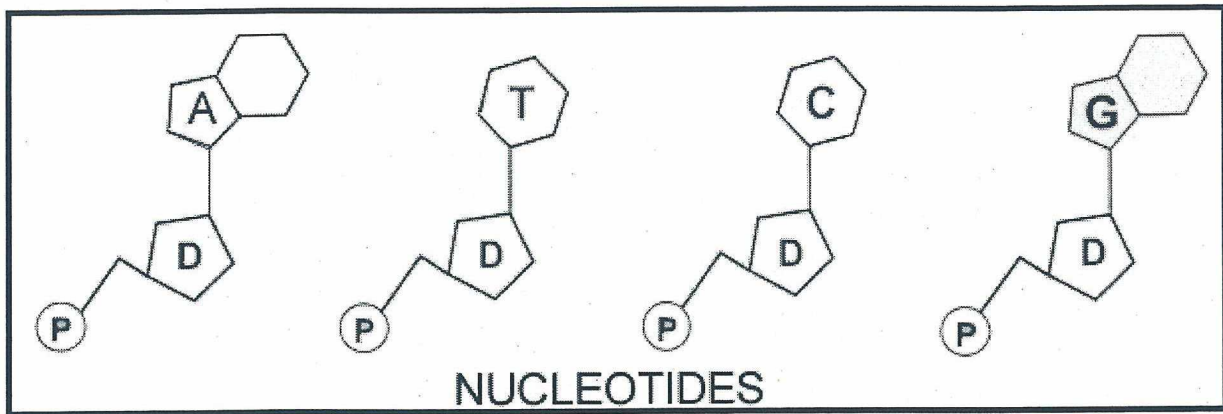
10. How do the bases bond together? A bonds with _____ G bonds with _____
11. The two purines in DNA are _____.
12. DNA is made of repeating units called _____
13. Why is RNA necessary to act as a messenger? Why can't the code be taken directly from the DNA?
14. Proteins are made where in the cell?
15. How do some cells become brain cells and others become skin cells, when the DNA in ALL the cells is exactly the same. In other words, if the instructions are exactly the same, how does one cell become a brain cell and another a skin cell?
16. Why is DNA called the "Blueprint of Life"?

Name _____

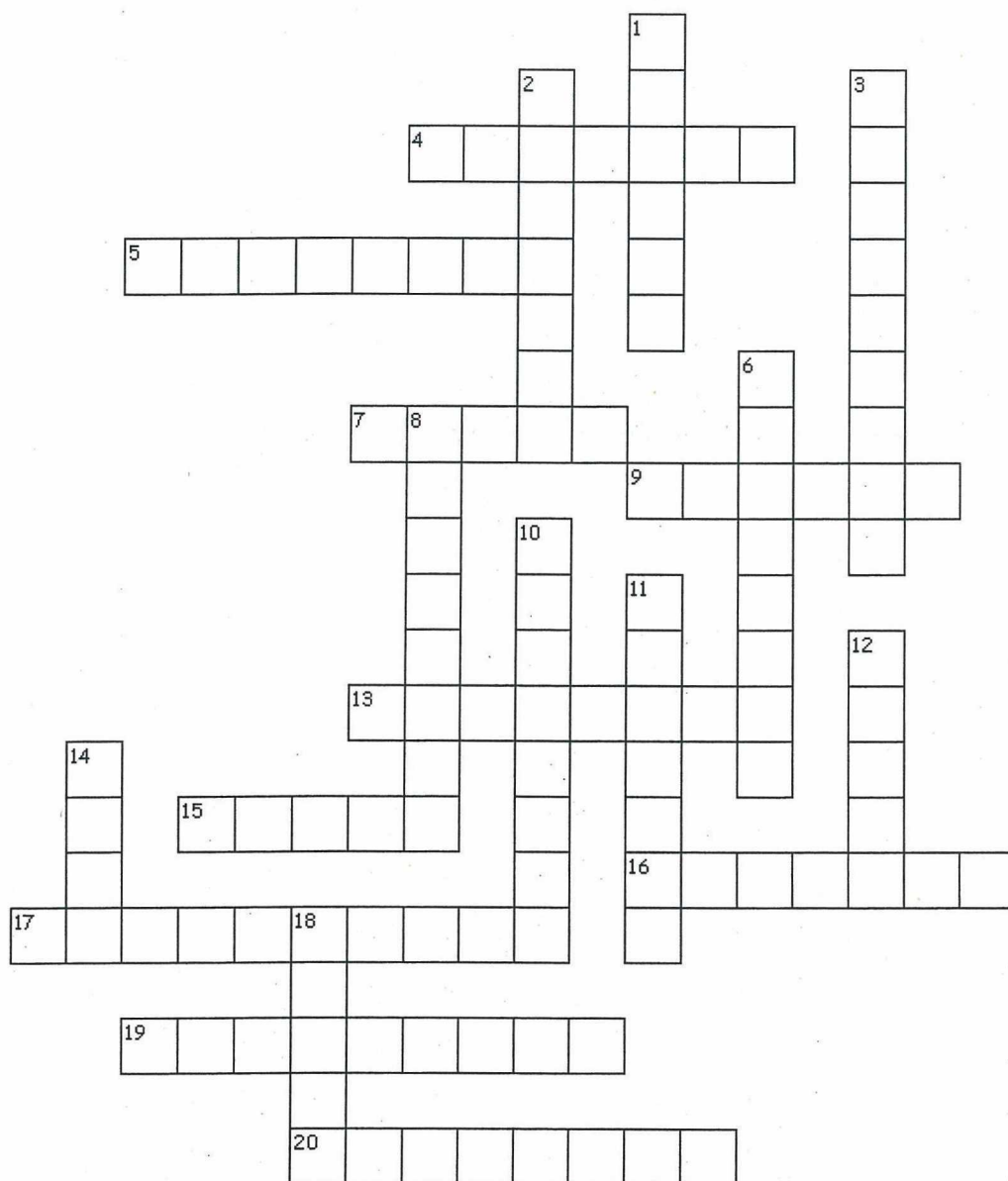
Color the images according to the instructions.

DNA - The Double Helix





CHEMICAL BONDING



Across

4. a covalent network substance that is very hard and shiny
5. a covalent network substance that will conduct electricity
7. carbon forms a covalent bond with other carbons in graphite
9. this element has an atomic number of 11.
13. the group one elements are very....
15. covalent molecular substances are usually in this state
16. made up of protons and neutrons
17. metals in the middle of the periodic table
19. a plus two positively charged metal
20. a type of chemical bond that involves sharing electrons

Down

1. found in the nucleus of an atom
2. the outer shell electrons
3. lightweight, shiny, malleable substance with valency of plus three
6. this gas is produced when a metal reacts with an acid
8. another name for the group seven elements
10. the negatively charged particle of an atom
11. ionic substance have high ... points
12. a group of gases is sometimes called this
14. carbon forms a covalent bond with other carbons in diamond
18. a type of bonding that involves transfer of electrons